I.22 · Construct a triangle from three lengths

To construct a triangle from three straight lines equal to three given [straight lines]. The two must be greater than the remaining one, however they are taken [because in every triangle, too, two sides are greater than the remaining side, however they are taken].

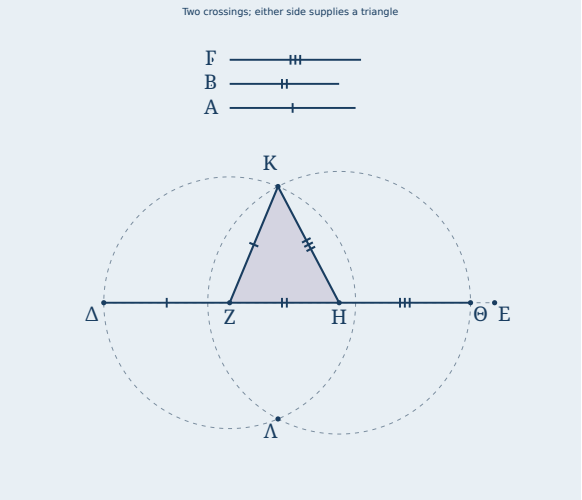


Plate at step 8: What was required to be made.

Two crossings; either side supplies a triangle

A, B and Γ name given lengths here; Z, H and K name the triangle's vertices. Copy lengths with I.2–3, then draw circles through available points. Flip the plate to choose the reflected vertex. For an unconstrained trial hinge and finer length controls, open the separate I.22 workshop.

Check every pairing before joining the circles

Two lengths	Remaining	Correspondence
A + B > Γ	387.00 > 216.00	Strict
A + Γ > B	423.00 > 180.00	Strict
B + Γ > A	396.00 > 207.00	Strict

ZK = ZΔ = A · HK = HΘ = Γ · ZH = B

The proposition

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας].	Ex tribus rectis, quae tribus datis aequales sunt, triangulum construere (oportet autem duas reliqua maiores esse quoquo modo coniunctas).	To construct a triangle from three straight lines equal to three given [straight lines]. The two must be greater than the remaining one, however they are taken [because in every triangle, too, two sides are greater than the remaining side, however they are taken].

1. Check all three pairings Given

Ἔστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ Α, Β, Γ, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν Α, Β τῆς Γ, αἱ δὲ Α, Γ τῆς Β, καὶ ἔτι αἱ Β, Γ τῆς Α· δεῖ δὴ ἐκ τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συστήσασθαι.	Sint tres datae rectae A, B, Γ, quarum duae reliqua maiores sint quoquo modo coniunctae, A + B > Γ, A + Γ > B, B + Γ > A. oportet igitur ex rectis aequalibus rectis A, B, Γ triangulum construere.	Let A, B and Γ be the three given straight lines, and let any two be greater than the remaining one: A and B greater than Γ, A and Γ greater than B, and also B and Γ greater than A. We must construct a triangle from lines equal to A, B and Γ.

ἔστωσαν / sint: ‘let them be’, plural. The genitive τῆς Γ follows the comparative μείζονες: ‘greater than Γ’. Here A, B and Γ name whole lengths, not the vertices of a triangle.

Reason: Check all three pairings The givens include three strict inequalities. The experiments deliberately allow failures, but Euclid's proof below applies only while all three hold.

Printed source: Greek pp. 52; Latin pp. 53.

2. Place the lengths consecutively Construct

Ἐκκείσθω τις εὐθεῖα ἡ ΔΕ πεπερασμένη μὲν κατὰ τὸ Δ ἄπειρος δὲ κατὰ τὸ Ε, καὶ κείσθω τῇ μὲν Α ἴση ἡ ΔΖ, τῇ δὲ Β ἴση ἡ ΖΗ, τῇ δὲ Γ ἴση ἡ ΗΘ.

sumatur recta ΔΕ terminata in Δ, uersus Ε autem infinita, et ponatur ΔΖ = Α, ΖΗ = Β, ΗΘ = Γ.

Set out a straight line ΔΕ, ending at Δ and extending without end towards Ε. Place ΔΖ equal to Α, ΖΗ equal to Β, and ΗΘ equal to Γ.

πεπερασμένη μὲν … ἄπειρος δὲ contrasts an endpoint on one side with no endpoint on the other. κείσθω / ponatur commands the placement. The line on the screen is necessarily only a finite part of that ray.

Reason: Place the lengths consecutively The preceding length-transfer and cut-off constructions (I.2–3) supply the three copies. These are reused results, not a new permission to carry an arbitrary compass opening. ΖΗ, the middle copy, becomes the base.

Printed source: Greek pp. 52, 54; Latin pp. 53, 55.

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3. Draw the first circle Construct

καὶ κέντρῳ μὲν τῷ Ζ, διαστήματι δὲ τῷ ΖΔ κύκλος γεγράφθω ὁ ΔΚΛ·

et centro Ζ radio autem ΖΔ circulus describatur ΔΚΛ.

With centre Ζ and distance ΖΔ, let the circle ΔΚΛ be drawn.

κέντρῳ … διαστήματι are datives specifying centre and distance; Latin uses centro … radio. γεγράφθω is a perfect passive imperative: ‘let it have been drawn’. Κ and Λ name the crossings anticipated in the successful construction.

Reason: Draw the first circle Postulate 3 supplies the circle at an available centre through an available point. Every point of this circumference is the required distance Α from Ζ.

Printed source: Greek pp. 54; Latin pp. 55.

4. Draw the second; join at K Construct

πάλιν κέντρῳ μὲν τῷ Η, διαστήματι δὲ τῷ ΗΘ κύκλος γεγράφθω ὁ ΚΛΘ, καὶ ἐπεζεύχθωσαν αἱ ΚΖ, ΚΗ· λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συνέσταται τὸ ΚΖΗ.

rursus centro Η radio autem ΗΘ circulus describatur ΚΛΘ, et ducantur ΚΖ, ΚΗ. dico, ex tribus rectis aequalibus rectis Α, Β, Γ triangulum constructum esse ΚΖΗ.

Again, with centre Η and distance ΗΘ, let the circle ΚΛΘ be drawn, and join ΚΖ and ΚΗ. I say that triangle ΚΖΗ has been constructed from three straight lines equal to Α, Β and Γ.

πάλιν / rursus repeats the operation with another centre. ἐπεζεύχθωσαν / ducantur is plural: two lines are to be joined. συνέσταται says that the triangle has now been constructed. The interface separates drawing the second circle from joining the vertex.

Reason: Draw the second; join at K Euclid proceeds directly to joining K. The existence of the common point needs justification beyond merely drawing two circles. The modern inequality and coordinate explanation below supplies a model, not a missing sentence in the ancient proof.

Printed source: Greek pp. 54; Latin pp. 55.

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5. The first radius supplies A Prove

Ἐπεὶ γὰρ τὸ Ζ σημεῖον κέντρον ἐστὶ τοῦ ΔΚΛ κύκλου, ἴση ἐστὶν ἡ ΖΔ τῇ ΖΚ· ἀλλὰ ἡ ΖΔ τῇ Α ἐστιν ἴση· καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση.

nam quoniam Ζ punctum centrum est circuli ΔΚΛ, erit ΖΔ = ΖΚ; uerum ΖΔ = Α; quare etiam ΚΖ = Α.

Since point Ζ is the centre of circle ΔΚΛ, ΖΔ equals ΖΚ. But ΖΔ equals Α; therefore ΚΖ also equals Α.

ἐπεὶ γάρ / nam quoniam begins the reason. ἀλλά / uerum supplies the previously established equality. ἄρα / quare signals the conclusion obtained by connecting them.

Reason: The first radius supplies A Radii of one circle are equal (definition 15 in this course). Common notion 1 carries the equality through the copied length ΖΔ. A measurement is a check of this drawing, not Euclid's reason.

Printed source: Greek pp. 54; Latin pp. 55.

6. The other radius supplies Γ Prove

πάλιν, ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ΛΚΘ κύκλου, ἴση ἐστὶν ἡ ΗΘ τῇ ΗΚ· ἀλλὰ ἡ ΗΘ τῇ Γ ἐστιν ἴση· καὶ ἡ ΚΗ ἄρα τῇ Γ ἐστιν ἴση.

rursus quoniam Η punctum centrum est circuli ΛΚΘ, erit ΗΘ = ΗΚ; uerum ΗΘ = Γ; quare etiam ΚΗ = Γ.

Again, since point Η is the centre of circle ΛΚΘ, ΗΘ equals ΗΚ. But ΗΘ equals Γ; therefore ΚΗ also equals Γ.

The same reasoning returns with Η as centre and Γ as the given length. The circle is named ΛΚΘ here, rather than ΚΛΘ as in the construction; the transcription retains that change of order.

Reason: The other radius supplies Γ The second circumference certifies the second copied radius. A point on the first circle alone is not enough: this same Κ must also lie on the circle at Η.

Printed source: Greek pp. 54; Latin pp. 55.

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7. Keep the middle copy as the base Prove

ἐστὶ δὲ καὶ ἡ ΖΗ τῇ Β ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΚΖ, ΖΗ, ΗΚ τρισὶ ταῖς Α, Β, Γ ἴσαι εἰσίν.

et praeterea ΖΗ = Β. itaque tres rectae ΚΖ, ΖΗ, ΗΚ tribus Α, Β, Γ aequales sunt.

And ΖΗ also equals Β. The three straight lines ΚΖ, ΖΗ and ΗΚ therefore equal the three Α, Β and Γ.

τρεῖς is nominative, the three lines being discussed; τρισί is dative, the three lengths to which they are equal. The Latin likewise contrasts tres and tribus.

Reason: Keep the middle copy as the base ΖΗ was copied before either circle was drawn. The two circles supply the other two sides; they do not change the base to make the triangle fit.

Printed source: Greek pp. 54; Latin pp. 55.

8. What was required to be made Conclude

Ἐκ τριῶν ἄρα εὐθειῶν τῶν ΚΖ, ΖΗ, ΗΚ, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς Α, Β, Γ, τρίγωνον συνέσταται τὸ ΚΖΗ· ὅπερ ἔδει ποιῆσαι.	Ergo ex tribus rectis KZ, ZH, HK, quae tribus datis rectis A, B, Γ aequales sunt, triangulus constructus est KZH; quod oportebat fieri.	Thus triangle KZH has been constructed from the three straight lines KZ, ZH and HK, equal to the three given straight lines A, B and Γ: which was required to be made.

ὅπερ ἔδει ποιῆσαι / quod oportebat fieri closes a construction. Compare ὅπερ ἔδει δεῖξαι, ‘what was required to be shown’, at the end of I.20's theorem.

Reason: What was required to be made The conclusion belongs to the successful case with the original three strict inequalities. At a tangent there is only a straight configuration; when the circles miss, there is no common vertex at all.

Printed source: Greek pp. 54; Latin pp. 55.

From a necessary condition to a usable construction

I.20 established what the sides of an existing triangle must satisfy. I.22 now turns suitable lengths into a triangle. Try a limiting equality: a tangent point lies on the line of centres and does not make a genuine triangle. The separate I.22 workshop includes a trial hinge and exact tenths. Continue to I.23 to use this earned construction to copy an angle.

Definitions and starting permissions used in this proof

Postulate 3 · γ′ / III · Draw a circle

Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.	Et ut quouis centro radioque circulus describatur.	And that a circle can be drawn with any centre and distance.

κέντρῳ and διαστήματι are datives; Latin centro and radio are ablatives. The two languages attach the circle-drawing operation to a centre and a distance. Compare the fuller command Κέντρῳ … διαστήματι … γεγράφθω in I.1.

Printed source: Greek pp. 8; Latin pp. 9.

Postulate 1 · α′ / I · Join two points

Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.	Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.	Let it be postulated that a straight line can be drawn from any point to any point.

Ἠιτήσθω / Postuletur means ‘let it be requested / postulated’. The Greek then uses an infinitive, ἀγαγεῖν (‘to draw’); Latin uses ut … ducatur, ‘that … be drawn’. The following postulates continue the opening request.

Printed source: Greek pp. 8; Latin pp. 9.

Definition 15 · ιε′ / XV · A circle and its equal radii

Κύκλος ἐστὶ σχῆμα ἐπίπεδον ὑπὸ μιᾶς γραμμῆς περιεχόμενον [ἢ καλεῖται περιφέρεια], πρὸς ἣν ἀφ’ ἑνὸς σημείου τῶν ἐντὸς τοῦ σχήματος κειμένων πᾶσαι αἱ προσπίπτουσαι εὐθεῖαι [πρὸς τὴν τοῦ κύκλου περιφέρειαν] ἴσαι ἀλλήλαις εἰσίν.	Circulus est figura plana una linea comprehensa, ad quam quae ab uno puncto intra figuram posito educuntur rectae omnes aequales sunt.	A circle is a plane figure enclosed by one line [called the circumference], to which all the straight lines falling from one point within the figure [to the circumference of the circle] are equal to one another.

The bracketed clauses are printed in Heiberg’s Greek and omitted from his Latin. They are retained here. πᾶσαι … εὐθεῖαι means ‘all the straight lines’; the statement supplies equal radii, not just two radii that happen to look alike.

Printed source: Greek pp. 4; Latin pp. 5.

Common notion 1 · α′ / I · Equality through a shared equal

Τὰ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα.	Quae eidem aequalia sunt, etiam inter se aequalia sunt.	Things equal to the same thing are also equal to one another.

τῷ αὐτῷ and eidem mean ‘to the same [thing]’. The Greek neuter plural τὰ … ἴσα takes the singular ἐστίν, a normal Greek construction. Latin uses plural sunt.

Printed source: Greek pp. 10; Latin pp. 11.

Greek and Latin: Euclidis opera omnia, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1883, printed pp. 2–57. Public domain. University of Toronto scan. English, reading notes and redrawn coordinate diagrams: Stoicheia (2026), MIT. Greek ϑ is normalized to θ; line breaks and

inline references are omitted. Latin u/v spelling is retained. Heiberg’s square brackets in I.1, I.4, I.8, I.13 and I.15 are retained. Printed angle and triangle symbols are represented as ∠ and △. The English is a reading translation, including explicit explanations of implied intersections.

The diagram is numerical evidence, not a formal proof. Length units and orientation are modern teaching choices.

The printed evidence

Source pages from Heiberg’s 1883 edition, including the passages and foundations used in this saved reading. Every image and the bundled font were checked against this edition before saving. The apparatus is available for inspection; it is not fully transcribed. Where the original scan is cropped, the page caption identifies that limit.

ἐξῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν ἐστι, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται, ἐφ' ἣν ἐφέστηκεν.

ια΄. Ἀμβλεῖα γωνία ἐστὶν ἡ μείζων ὀρθῆς.

ιβ΄. Ὀξεῖα δὲ ἡ ἐλάσσων ὀρθῆς.

ιγ΄. Ὅρος ἐστίν, ὅ τινός ἐστι πέρας.

ιδ΄. Σχῆμά ἐστι τὸ ὑπό τινος ἢ τινων ὅρων περιεχόμενον.

ιε΄. Κύκλος ἐστὶ σχῆμα ἐπίπεδον ὑπὸ μιᾶς γραμμῆς περιεχόμενον [ἣ καλεῖται περιφέρεια], πρὸς ἣν ἀφ' ἑνὸς σημείου τῶν ἐντὸς τοῦ σχήματος κειμένων πᾶσαι αἱ προσπίπτουσαι εὐθεῖαι [πρὸς τὴν τοῦ κύκλου περιφέρειαν] ἴσαι ἀλλήλαις εἰσίν.

ις΄. Κέντρον δὲ τοῦ κύκλου τὸ σημεῖον καλεῖται.

ιζ΄. Διάμετρος δὲ τοῦ κύκλου ἐστὶν εὐθεῖά τις διὰ τοῦ κέντρου ἠγμένη καὶ περατουμένη ἐφ' ἑκάτερα τὰ μέρη ὑπὸ τῆς τοῦ κύκλου περιφερείας, ἥτις καὶ δίχα τέμνει τὸν κύκλον.

ιη΄. Ἡμικύκλιον δέ ἐστι τὸ περιεχόμενον σχῆμα ὑπό τε τῆς διαμέτρου καὶ τῆς ἀπολαμβανομένης ὑπ'

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11. Hero def. 21. Ammonius in categ. p. 58. Psellus p. 36. Martianus Capella VI, 710. Boetius p. 374, 18. 12. Hero def. 20. Ammonius l. c. Psellus l. c. Martianus Capella l. c. Boetius p. 374, 19. 13. Philoponus in Aristot. de anima fol. a 2. Martianus Capella VI, 710. Boetius p. 374, 22. 14. Hero def. 25. Schol. in Hermog. VII² p. 903. cfr. Philop. ad Aristot. de anim. h. 7. Martianus Capella VI, 710. Boetius p. 374, 21. 15. Hero def. 29. Taurus apud Philop. in Proclum VI, 21. Sextus Emp. p. 719, 16. Philopon. in anal. II fol. 28^{v}, cfr. fol. 4^{v}, 9^{v}, 29^{r}, 53^{r}. Psellus p. 38. Martianus Capella VI, 710. Boetius p. 375, 3. 16. Psellus p. 38. Martianus Capella VI, 711. Boetius p. 375, 6. 17. Hero def. 30. Psellus p. 38. Martianus Capella VI, 711. Boetius p. 375, 7. 18. Hero def. 31. Mart. Capella VI, 711. Boetius p. 375, 12.

angulos deinceps positos inter se aequales efficit, rectus est uterque angulus aequalis, et recta linea erecta perpendicularis adpellatur ad eam, super quam erecta est.

XI. Obtusus angulus est, qui maior est recto.

XII. Acutus uero, qui minor est recto.

XIII. Terminus est, quod alicuius rei extremum est.

XIV. Figura est, quod aliquo uel aliquibus terminis comprehenditur.

XV. Circulus est figura plana una linea comprehensa, ad quam quae ab uno puncto intra figuram posito educuntur rectae omnes aequales sunt.

XVI. Centrum autem circuli punctum illud adpellatur.

XVII. Diametrus autem circuli recta quaedam est linea per centrum ducta et terminata utrimque ambitu circuli, quae quidem linea circulum in duas partes aequales diuidit.

XVIII. Semicirculus autem ea est figura, quae

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1. *ὀρθὴ ἐστιν ἑκατέρα* omisso *ἐστι* lin. 2 BFV, Simplicius, Philoponus in anal. II p. 65, Psellus. scripturam receptam praebent Pbp, Proclus, Hero, Ammonius, Philoponus in phys. i IIII. cfr. prop. 11, 12. 2. *ἴσων*] om. Ammonius, Philoponus in phys. l. c., Psellus, Martianus Capella, Campanus. *εὐθεῖα*] *γραμμή* Proclus, BV; om. Ammonius. Deff. XI—XII permutant Hero et Ammonius. 6. *ιγ΄*] *ιδ΄* V et sic deinceps. Deff. XIII—XIV permutat Boetius. 7. *ἐστι*] *δέ* Fbp. 10. *ἣ καλεῖται περιφέρεια*] om. Proclus, Taurus, Sextus Emp., Philoponus, Boetius; habent praeter codd. Hero, Psellus, Capella, Campanus. 12. *προπίπτουσαι* b, corr. m. 2. *πρὸς τὴν τοῦ κύκλου περιφέρειαν*] om. Proclus, Taurus, Hero, Sextus Emp., Psellus, Capella, Boetius; habent codd. (in b erasa sunt), Philoponus, Campanus. 13. *εἰσίν*] PF, *εἰσί* uulgo. 19. *ἐστιν* PF. 20. *τε*] om. B. *καί*] *τε καί* B. *ὑπολαμβανομένης* B.

οὔτε ὀρθογώνιον· τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλείσθω.

κγ΄. Παράλληλοί εἰσιν εὐθεῖαι, αἵτινες ἐν τῷ αὐτῷ ἐπιπέδῳ οὖσαι καὶ ἐκβαλλόμεναι εἰς ἄπειρον ἐφ' ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαις.

Αἰτήματα.

α΄. Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.

β΄. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ' εὐθείας ἐκβαλεῖν.

γ΄. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.

δ΄. Καὶ πάσας τὰς ὀρθὰς γωνίας ἴσας ἀλλήλαις εἶναι.

ε΄. Καὶ ἐὰν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίας δύο ὀρθῶν ἐλάσσονας ποιῇ, ἐκβαλλομένας τὰς δύο εὐθείας ἐπ' ἄπειρον συμπίπτειν, ἐφ' ἃ μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονες.

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23. Hero def. 71. Philoponus in anal. II fol. 18^{v}. Psellus p. 35. Martianus Capella VI, 712. Boetius p. 376, 23. αἰτ. 1—5. Martianus Capella VI, 722. Boetius p. 377, 4. Aspasius apud Simplicium in Arist. de coelo fol. 149: τὰ πέντε αἰτήματα. 1. Philop. in anal. II fol. 9^{v}. 10. 29. 2. Simplicius in phys. fol. 119. 3. Philop. in anal. II fol. 10. 29. 4. Id. ibid. fol. 10. 5. Id. ib. fol. 10. 29. Proclus p. 364, 14.

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1. τετράγωνα B. 2. τραπέζεια b. Def. 21 uulgo in 3, def. 22 in 5 diuidunt. 3. παράλληλοι δέ B. εὐθεῖαί εἰσιν Proclus, Psellus. 4. ἐς V. 5. συμπίπτειν P. ἀλλήλαις om. F. 6. αἰτήματα πέντε V, αἰτ. ἐστι πέντε BF, b m. 2. Numeros om. F. 9. ἐπ' εὐθείας κατὰ τὸ συνεχές PBFbp$_1$

liqua autem praeter haec quadrilatera trapezia adpellentur.

XXIII. Parallelae sunt lineae, quae in eodem plano positae et in utramque partem productae in infinitum in neutra parte concurrunt.

Postulata.

I. Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.

II. Et ut recta linea terminata in directum educatur in continuum.

III. Et ut quouis centro radioque circulus describatur.

IV. Et omnes rectos angulos inter se aequales esse.

V. Et, si in duas lineas rectas recta incidens angulos interiores et ad eandem partem duobus rectis minores effecerit, rectas illas in infinitum productas concurrere ad eandem partem, in qua sint anguli duobus rectis minores.

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receptum ordinem tuentur V, Proclus, Simplicius, Capella, Boetius, Campanus. 10. *ἐκβάλλειν* V. 11. *γράφεσθαι*] codd. omnes et Philoponus; *γράψαι* ex Proclo recepit August. 13. *ἀλλήλαις*] om. V. 15. *εὐθειά τις* P. 17. *ἐλάττονας* Proclus p. 191, 18 (non p. 364). *τὰς δύο*] PBVbp, *δύο* om. F, Proclus bis, Martianus Capella, Boetius, fort. recte. 18. *συμπίπτειν τὰς εὐθείας ἐκβαλλομένας ἐφ'* Proclus p. 364. *συμπίπτειν ἀλλήλαις* PV (*ἀλλήλαις* corr. ex *ἀλλήλας* P). 19. *ἐλάσσονες*] Pp, Proclus p. 364; *ἐλάττονες* uulgo. Dein add. *γωνίαι* FBVb, Philoponus; om. Proclus bis et Pp. In ed. Basil. et apud Gregorium *αἴτ.* 4—5 inter communes notiones (10—11) leguntur (*πᾶσαι αἱ ὀρθαὶ γωνίαι ἴσαι .. εἰσί; ἐκβαλλόμεναι αἱ .. εὐθεῖαι .. συμπεσοῦνται*). Post *αἴτ.* 5 in PF et V m. 2 et apud Campanum sequitur: *καὶ δύο εὐθείας χωρίον μὴ περιέχειν.*

Κοιναὶ ἔννοιαι.

α΄. Τὰ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα.

β΄. Καὶ ἐὰν ἴσοις ἴσα προστεθῇ, τὰ ὅλα ἐστὶν ἴσα.

γ΄. Καὶ ἐὰν ἀπὸ ἴσων ἴσα ἀφαιρεθῇ, τὰ καταλειπόμενά ἐστιν ἴσα.

[δ΄. Καὶ ἐὰν ἀνίσοις ἴσα προστεθῇ, τὰ ὅλα ἐστὶν ἄνισα.

ε΄. Καὶ τὰ τοῦ αὐτοῦ διπλάσια ἴσα ἀλλήλοις ἐστίν.

ϛ΄. Καὶ τὰ τοῦ αὐτοῦ ἡμίση ἴσα ἀλλήλοις ἐστίν.]

ζ΄. Καὶ τὰ ἐφαρμόζοντα ἐπ᾽ ἀλλήλα ἴσα ἀλλήλοις ἐστίν.

η΄. Καὶ τὸ ὅλον τοῦ μέρους μεῖζόν [ἐστιν].

[θ΄. Καὶ δύο εὐθεῖαι χωρίον οὐ περιέχουσιν.]

α΄.

Ἐπὶ τῆς δοθείσης εὐθείας πεπερασμένης τρίγωνον ἰσόπλευρον συστήσασθαι.

Ἔστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB.

Δεῖ δὴ ἐπὶ τῆς AB εὐθείας τρίγωνον ἰσόπλευρον συστήσασθαι.

Κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ AB κύκλος

Κοιν. ἔνν. 1—3. Martianus Capella VI, 723. 1. Philop. in anal. II fol. 5. Boetius p. 378, 1. 2. Boetius p. 378, 5. 3. Philop. l. c. Boetius p. 378, 3. 4. Eutocius in Archim. III p. 254, 27. 7. Philop. in anal. II fol. 5. Boetius p. 378, 7. prop. I. Alexander Aphrod. in anal. I fol. 8^{r}, in top. p. 11. Themistius phys. paraphr. fol. 35^{v}. Simplicius in phys. fol. 119. Proclus p. 102, 14. 223, 22, Philop. in anal. II fol. 4^{v}. Martianus Capella VI, 724. Boetius p. 380, 2 [p. 390, 6—25]. Proclus p. 208—10 liberius proposit. repetit totam.

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1. *ἀξιώματα* Proclus p. 193. *κοιν. ἔνν. αἵδε* BFV. numeros om. PBF. 3. *ἴσα ἴσοις* Proclus. *ἴσα ἐστίν* Proclus. 4. *ἀπὸ ἴσων ἴσα*] *ἴσων* Proclus. 5. *ἴσα ἐστίν* Proclus. *αἴτ.* 4 ex commentario Pappi irrepsisse uidetur; u. Proclus

Communes animi conceptiones.

I. Quae eidem aequalia sunt, etiam inter se aequalia sunt.

II. Et, si aequalibus aequalia adduntur, tota aequalia sunt.

III. Et, si ab aequalibus aequalia subtrahuntur, reliqua sunt aequalia.

VII. Et quae inter se congruunt, aequalia sunt.

VIII. Et totum parte maius est.

I.

In data recta terminata triangulum aequilaterum construere.

Sit data recta terminata AB. oportet igitur in recta AB terminata triangulum aequilaterum construere.

centro A et radio AB circulus describatur $B\Gamma\Delta$,

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p. 197, 6 sq.; in omnibus codicibus legitur; quare iam ante Theonem receptum erat (P); om. Martianus Capella et Boetius. Ante *αἴτ.* 5 uulgo in codd. et edd. legitur: *καὶ ἐὰν ἀπὸ ἀνίσων ἴσα ἀφαιρεθῇ, τὰ λοιπά ἐστιν ἄνισα*; om. B, mg. Fb, in ras. postea additum p; non agnoscunt Proclus (cfr. p. 198, 3), Capella, Boetius. *αἴτ.* 5—6 reiicit Proclus p. 196, 25, om. Capella et Boetius. *αἴτ.* 7—8 permutat Proclus p. 193, qui ea diserte contra Heronem sola *αἴτ.* 1—3 agnoscentem Euclidi uindicat p. 196, 17; om. Capella; *αἴτ.* 8 etiam Boetius om. *αἴτ.* 9 om. Capella, Boetius, Proclus, qui diserte id improbat p. 184, 8. 196, 23. Hoc loco habent Vbp; cfr. Philop. ad phys. fol. 10; *καὶ δύο εὐθείας χωρίον μὴ περιέχειν* B; de ceteris u. ad p. 8, 19. 8. *ἐστίν*] PF, *ἐστί* uulgo; comp. b; item lin. 9. 10. 10. *ἐπ' ἄλληλα*] om. Proclus. *ἐστίν*] *εἰσί* B. 11. *ἐστίν*] om. Proclus; comp. b; ///*αι* F, *εἶναι* P. 17. *εὐθείας*] om. BFbp. *εὐθείας πεπερασμένης* P. 19. *μέν*] om. bp. *καὶ διαστηματι* Bp. *δέ* om. BFbp.

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ ΓΔΕ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΓΕΔ. διὰ ταὐτὰ τοίνυν καὶ τοῦ ΑΒΕ τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΕΒ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ. ἀλλὰ τῆς ὑπὸ ΓΕΒ μείζων ἐδείχθη ἡ ὑπὸ ΒΔΓ· πολλῷ ἄρα ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.

Ἐὰν ἄρα τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μέν εἰσιν, μείζονα δὲ γωνίαν περιέχουσιν· ὅπερ ἔδει δεῖξαι.

κβ'.

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας].

Ἔστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ Α, Β, Γ, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν Α, Β τῆς Γ, αἱ δὲ Α, Γ τῆς Β, καὶ ἔτι αἱ Β, Γ τῆς Α· δεῖ δὴ ἐκ τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συστήσασθαι.

Ἐκκείσθω τις εὐθεῖα ἡ ΔΕ πεπερασμένη μὲν κατὰ

XXII. Proclus p. 102, 16. Eutocius in Apollonium p. 10. Boetius p. 382, 1 (male). partem demonstrationis habet Proclus p. 330 sq.

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2. ἐντός] ἐν- in ras. b. ἐστίν] PF; ἐστί uulgo. ΓΔΕ] e corr. F m. 2; mutat. in ΓΕΔ V. ἄρα] supra F. 3.

rursus quoniam in quouis triangulo angulus extrinsecus positus maior est angulo interiore et opposito [prop. XVI], in triangulo $\Gamma\Delta E$ erit $\angle B\Delta\Gamma > \Gamma E\Delta$. eadem de causa igitur etiam in triangulo ABE erit $\angle\Gamma EB > BA\Gamma$. uerum demonstratum est $\angle B\Delta\Gamma > \Gamma EB$. multo igitur magis $B\Delta\Gamma > BA\Gamma$.

Ergo si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent; quod erat demonstrandum.

XXII.

Ex tribus rectis, quae tribus datis aequales sunt, triangulum construere (oportet autem duas reliqua maiores esse quoquo modo coniunctas [prop. XX]).

Sint tres datae rectae A, B, Γ, quarum duae reliqua maiores sint quoquo modo coniunctae, $A + B > \Gamma$, $A + \Gamma > B$, $B + \Gamma > A$. oportet igitur ex rectis aequalibus rectis A, B, Γ triangulum construere.

sumatur[1]) recta ΔE terminata in Δ, uersus E au-

1) Proclum non ipsa uerba Euclidis citare, adparet. cfr. idem p. 102, 19. Augustum perperam post $K\Lambda\Theta$ p. 54, 5. suppleuisse: *καὶ τεμνέτωσαν ἀλλήλους οἱ κύκλοι κατὰ τὸ K*, demonstraui „Studien“ p. 185.

$B\Delta\Gamma$] Δ in ras. F. *ἐστιν* PV. 4. $\Gamma E\Delta$] eras. F. *ταὐτά*] *τὰ αὐτά* F; *ταῦτα* Vbp. 5. *ἐστίν* P, ut lin. 7. 6. *ἀλλὰ καὶ τῆς* F. 7. $B\Delta\Gamma$] (alt.) $B\Delta$ in ras. sunt V. 12. *εἰσιν*] P; *εἰσι* uulgo. 15. *αἳ εἰσιν τρισὶ ταῖς δοθείσαις εὐθείαις ἴσαι* Proclus p. 329; sed p. 102: *αἳ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις.* 16. *εὐθείαις*] om. b; m. rec. P; supra p; mg. m. 2 V; om. Eutocius. 17. *δέ*] Proclus, Eutocius; *δή* codd. *τάς*] corr. ex *ταῖς* F. *δύο*] *β* b. 18. *διὰ τὸ* — 20. *μεταλαμβανομένας*] omnes codd., Boetius; om. Proclus, Campanus; contra Eutocius ea habuisse uidetur. 21. *τρεῖς*] om. p.

τὸ Δ ἄπειρος δὲ κατὰ τὸ Ε, καὶ κείσθω τῇ μὲν Α ἴση ἡ ΔΖ, τῇ δὲ Β ἴση ἡ ΖΗ, τῇ δὲ Γ ἴση ἡ ΗΘ· καὶ κέντρῳ μὲν τῷ Ζ, διαστήματι δὲ τῷ ΖΔ κύκλος γεγράφθω ὁ ΔΚΛ· πάλιν κέντρῳ μὲν τῷ Η, διαστήματι δὲ τῷ ΗΘ κύκλος γεγράφθω ὁ ΚΛΘ, καὶ ἐπεζεύχθωσαν αἱ ΚΖ, ΚΗ· λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συνέσταται τὸ ΚΖΗ.

Ἐπεὶ γὰρ τὸ Ζ σημεῖον κέντρον ἐστὶ τοῦ ΔΚΛ κύκλου, ἴση ἐστὶν ἡ ΖΔ τῇ ΖΚ· ἀλλὰ ἡ ΖΔ τῇ Α ἐστιν ἴση. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση. πάλιν, ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ΛΚΘ κύκλου, ἴση ἐστὶν ἡ ΗΘ τῇ ΗΚ· ἀλλὰ ἡ ΗΘ τῇ Γ ἐστιν ἴση· καὶ ἡ ΚΗ ἄρα τῇ Γ ἐστιν ἴση. ἐστὶ δὲ καὶ ἡ ΖΗ τῇ Β ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΚΖ, ΖΗ, ΗΚ τρισὶ ταῖς Α, Β, Γ ἴσαι εἰσίν.

Ἐκ τριῶν ἄρα εὐθειῶν τῶν ΚΖ, ΖΗ, ΗΚ, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς Α, Β, Γ, τρίγωνον συνέσταται τὸ ΚΖΗ· ὅπερ ἔδει ποιῆσαι.

κγ'.

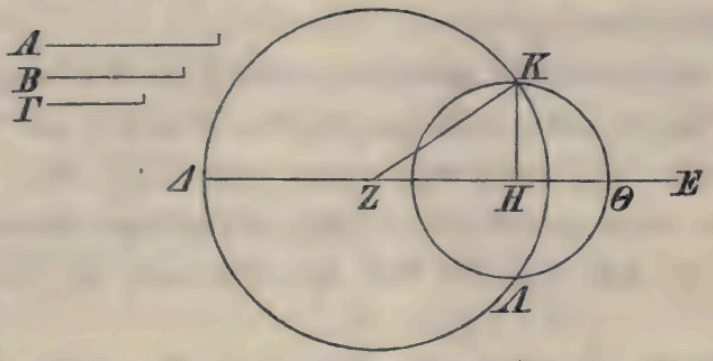
Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

XXIII. Boetius p. 382, 5.

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1. τῇ] postea insertum m. 1 V. 2. ἡ] (tert.) m. rec. P. 3. μέν] om. b, Proclus. 4. καὶ πάλιν V, Proclus. μέν] om. V, Proclus. διαστήματι δέ] καὶ διαστήματι P. 7. συνέστηκε V; συνίσταται p. τό] corr. ex τῷ b. 8. γάρ] οὖν P. ἐστίν P. 9. ΖΔ] ΔΖ F. ἀλλ F. ΖΔ] ΔΖ V (ante Δ ras., Ζ mg. m. 2). 10. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση] mg. m. 2 V. 11. ἐστίν Bb. ΔΚΘ] ΚΛΘ P, et in ras. V. 12. ἀλλ' F. 13. ΚΗ] corr. ex ΚΘ m. 2 P. 14. ΗΚ BF. ἐστιν ἴση] mg. m. 2 V. ἐστὶν δέ P. 16. τῶν]

tem infinita, et ponatur $\Delta Z = A$, $ZH = B$, $H\Theta = \Gamma$. et centro Z radio autem $Z\Delta$ circulus describatur $\Delta K\Lambda$. rursus centro H radio autem $H\Theta$ circulus describatur $K\Lambda\Theta$, et ducantur KZ, KH. dico, ex tribus rectis aequalibus rectis A, B, Γ triangulum constructum esse KZH.



nam quoniam Z punctum centrum est circuli $\Delta K\Lambda$, erit $Z\Delta = ZK$; uerum $Z\Delta = A$; quare etiam $KZ = A$ [κ. $\check{\epsilon}\nu\nu$. 1].[1]) rursus quoniam H punctum centrum est circuli $\Lambda K\Theta$, erit $H\Theta = HK$; uerum $H\Theta = \Gamma$; quare etiam $KH = \Gamma$. et praeterea $ZH = B$. itaque tres rectae KZ, ZH, HK tribus A, B, Γ aequales sunt.

Ergo ex tribus rectis KZ, ZH, HK, quae tribus datis rectis A, B, Γ aequales sunt, triangulus constructus est KZH; quod oportebat fieri.

XXIII.

Ad datam rectam et punctum in ea datum angulum rectilineum dato angulo rectilineo aequalem construere.

1) Cfr. Alexander Aphrod. in anal. I fol. 8. Studien p. 195.

τοῦ F. 17. τρισί] om. F. Γ] om. V. 18. συνίσταται p. 21. εὐθυγράμμῳ γωνίᾳ Proclus.